Algebraik ifodalarni ko'paytuvchilarga ajratish

2-test (qiyinroq) · 20 ta savol

Ism-familiya: _____ Sinf: _____ Sana: _____

Har savolda bitta to'g'ri javobni belgilang.

---

1. $24x^2y - 36xy^2$ **ni ko'paytuvchilarga to'liq ajrating.**

A) $12xy(2x - 3y)$ **B)** $6xy(4x - 6y)$

C) $12xy(2x - 3)$ **D)** $12x^2y(2 - 3y)$

---

2. **Ko'paytuvchilarga ajrating:** $-15a - 25ab$

A) $-5a(3 + 5b)$ **B)** $-5a(3 - 5b)$

C) $5a(3 + 5b)$ **D)** $-5(3a - 5ab)$

---

3. **Ko'paytuvchilarga ajrating:** $3x(2a - b) - 4y(b - 2a)$

A) $(2a - b)(3x - 4y)$ **B)** $(b - 2a)(3x + 4y)$

C) $(2a - b)(3x + 4y)$ **D)** $(2a - b)(4y - 3x)$

---

4. **Ko'paytuvchilarga ajrating:** $(4xy + 8x) - (3y + 6)$

A) $(y + 2)(4x + 3)$ **B)** $(y - 2)(4x - 3)$

C) $(y + 2)(4x - 6)$ **D)** $(y + 2)(4x - 3)$

---

5. **Soddalashtiring va ko'paytuvchilarga to'liq ajrating:** $2(5x - 3) - 4(x - 6)$

A) $6(x + 7)$ **B)** $2(3x + 9)$ **C)** $6(x + 3)$ **D)** $6(x - 5)$

---

6. **Kalkulyatorsiz hisoblang:** $0{,}75 \cdot 48 + 48 \cdot 0{,}25$

A) 48 **B)** 12 **C)** $4{,}8$ **D)** 36

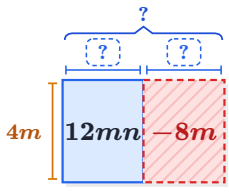
---

7. **Kalkulyatorsiz hisoblang:** $125 \cdot (-36) + 125 \cdot (-64)$

A) $-3\,500$ **B)** $12\,500$ **C)** $-12\,500$ **D)** $-1\,250$

---

8. **Rasmdagi to'g'ri to'rtburchakning yuzi** $12mn - 8m$**, bo'yi** $4m$**. Enini toping.**

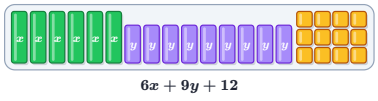


A) $3mn - 2$ **B)** $3n - 2$

C) $3n + 2$ **D)** $3n - 8$

---

9. **Rasmdagi barcha plitkalarni ko'pi bilan nechta teng guruhga ajratish mumkin va har bir guruhda nima bo'ladi? (Yashil —** x**, binafsha —** y**, sariq — 1.)**



A) 2 ta: $3x + 4y + 6$ **B)** 3 ta: $2x + 3y + 12$

C) 6 ta: $x + 3y + 2$ **D)** 3 ta: $2x + 3y + 4$

---

10. **Ko'paytuvchilarga to'liq ajrating:** $14x - 21y + 7$

A) $7(2x + 3y + 1)$ **B)** $7(2x - 3y + 1)$

C) $7(2x - 3y)$ **D)** $7(2x - 3y + 7)$

---

11. **Ko'paytuvchilarga to'liq ajrating:** $ab^2 - a^2b + ab$

A) $ab(b - a)$ **B)** $ab(b + a + 1)$

C) $ab(b - a + 1)$ **D)** $a(b^2 - ab + b)$

---

12. n **— natural son. Qaysi tasdiq har doim to'g'ri?**

A) $n^2 + n$ — juft son **B)** $n^2 + n$ — toq son

C) $n^2 + n$ — to'la kvadrat **D)** $n^2 + n$ 3 ga bo'linadi

---

13. **To'g'ri to'rtburchakning yuzi** $15ab + 10b$**, eni** $5b$**. Uning perimetrini toping.**

A) $3a + 5b + 2$ **B)** $6a + 10b + 4$

C) $6a + 10b + 2$ **D)** $6a + 4b + 10$

---

14. $c = 17$**,** $x = 58$**,** $y = 48$ **bo'lsa,** $cx - cy$ **ning qiymatini toping.**

A) $1\,700$ **B)** 170 **C)** 986 **D)** $1\,802$

---

15. $a = 4$**,** $x = 2{,}6$**,** $y = 1{,}2$ **bo'lsa,** $3ax + 6ay$ **ning qiymatini toping.**

A) $31{,}2$ **B)** 120 **C)** 60 **D)** $45{,}6$

---

16. **Maktab 36 ta futbol to'pi (har biri** x **so'm) va 54 ta voleybol to'pi (har biri** y **so'm) oldi: jami** $36x + 54y$ **so'm. To'plarni sinflarga teng bo'lish kerak: har bir sinfga futbol to'plari ham, voleybol to'plari ham bir xil sonda. To'plarni nechta sinfga bunday bo'lib BO'LMAYDI?**

A) 9 ta sinfga **B)** 12 ta sinfga

C) 6 ta sinfga **D)** 18 ta sinfga

---

17. **Qaysi ajratish to'g'ri?**

A) $-12a + 18b = -6(2a + 3b)$ **B)** $-12a + 18b = 6(2a - 3b)$

C) $-12a + 18b = -6(2a - 3b)$ **D)** $-12a + 18b = -6(-2a + 3b)$

---

18. **Ko'paytuvchilarga ajrating:** $2ax - 2ay + 3x - 3y$

A) $(x + y)(2a + 3)$ **B)** $(x - y)(2a - 3)$

C) $(x - y)(2a + 3)$ **D)** $(x - y)(2a + 3y)$

---

19. **Qaysi ifodaning ko'paytuvchilaridan biri** $(a - 2b)$**?**

A) $5a + 10b$ **B)** $5a - 10b$ **C)** $2a - 5b$ **D)** $10a - 5b$

---

20. **A to'g'ri to'rtburchakning tomonlari** $3x$ **va** $4y + 1$**, B niki —** $2x$ **va** $5y + 6$ **(**$x > 0$**).** y **ning qanday qiymatida ikkala yuz teng?**

A) 5 **B)** 9 **C)** 2 **D)** $4{,}5$

Javoblar

Algebraik ifodalarni ko'paytuvchilarga ajratish · 2-test (qiyinroq)

№	Javob	Qiymati
1	**A**	$12xy(2x-3y)$
2	**A**	$-5a(3+5b)$
3	**C**	$(2a-b)(3x+4y)$
4	**D**	$(y+2)(4x-3)$
5	**C**	$6(x+3)$
6	**A**	48
7	**C**	$-12\,500$
8	**B**	$3n-2$
9	**D**	3 ta: $2x+3y+4$
10	**B**	$7(2x-3y+1)$

№	Javob	Qiymati
11	**C**	$ab(b-a+1)$
12	**A**	n^2+n — juft son
13	**B**	$6a+10b+4$
14	**B**	170
15	**C**	60
16	**B**	12 ta sinfga
17	**C**	$-12a+18b=-6(2a-3b)$
18	**C**	$(x-y)(2a+3)$
19	**B**	$5a-10b$
20	**D**	$4{,}5$

Qisqa yechimlar

1. EKUB$(24;\ 36)=12$; x va y ikkala hadda bittadan bor: umumiy ko'paytuvchi $12xy$.
$24x^2y=12xy\cdot 2x$, $36xy^2=12xy\cdot 3y$. Javob: $12xy(2x-3y)$. ($6xy(4x-6y)$ — qavsda 2 qolgan.)

---

2. Minus bilan $5a$ ni chiqaramiz: $-15a=-5a\cdot 3$, $-25ab=-5a\cdot 5b$.
Qavs ichida ikkala ishora «+». Javob: $-5a(3+5b)$.

---

3. $b-2a=-(2a-b)$, shuning uchun $-4y(b-2a)=+4y(2a-b)$.
$3x(2a-b)+4y(2a-b)=(2a-b)(3x+4y)$. Javob: $(2a-b)(3x+4y)$.

---

4. $4xy+8x=4x(y+2)$, $3y+6=3(y+2)$.
$4x(y+2)-3(y+2)=(y+2)(4x-3)$. Javob: $(y+2)(4x-3)$.

---

5. Qavslarni ochamiz: $10x-6-4x+24=6x+18$ ($-4\cdot(-6)=+24$).
$6x+18=6(x+3)$. Javob: $6(x+3)$. ($2(3x+9)$ — tugallanmagan.)

---

6. 48 ni chiqaramiz: $48\cdot(0{,}75+0{,}25)=48\cdot 1$.
Javob: **48**.

---

7. $125\cdot(-36-64)=125\cdot(-100)$.
Javob: $-12\,500$.

---

8. $12mn=4m\cdot 3n$, $8m=4m\cdot 2$: $12mn-8m=4m(3n-2)$.
Javob: $3n-2$.

---

9. Plitkalar: $6x+9y+12$. EKUB$(6;\ 9;\ 12)=3$.
$6x+9y+12=3(2x+3y+4)$. 9 ta y 2 ga ham, 6 ga ham bo'linmaydi. Javob: **3 ta,** $2x+3y+4$.

---

10. $14x=7\cdot 2x$, $21y=7\cdot 3y$, $7=7\cdot 1$.
Uchinchi had o'rnida 1 qoladi. Javob: $7(2x-3y+1)$.

---

11. Uchala hadda a ham, b ham bor: umumiy ko'paytuvchi ab.
$ab^2=ab\cdot b$, $a^2b=ab\cdot a$, $ab=ab\cdot 1$. Javob: $ab(b-a+1)$.

---

12. $n^2 + n = n(n+1)$ — ketma-ket ikki son ko'paytmasi, ulardan biri juft.
$n = 1$ da 2 — 3 ga bo'linmaydi, to'la kvadrat ham emas. Javob: **juft son**.

---

13. 1) $15ab + 10b = 5b(3a + 2)$ — bo'yi $3a + 2$.
2) $P = 2(5b + 3a + 2) = 6a + 10b + 4$. Javob: $6a + 10b + 4$.

---

14. $cx - cy = c(x - y) = 17 \cdot (58 - 48) = 17 \cdot 10$.
Javob: **170**.

---

15. $3ax + 6ay = 3a(x + 2y)$.
$3 \cdot 4 \cdot (2{,}6 + 2{,}4) = 12 \cdot 5 = 60$. Javob: **60**. (45,6 — $x + y$ olinganda.)

---

16. Sinflar soni 36 ning ham, 54 ning ham bo'luvchisi bo'lishi kerak.
$36x + 54y = 6(6x + 9y) = 9(4x + 6y) = 18(2x + 3y)$. 12 esa 54 ning bo'luvchisi emas: $54 : 12 = 4{,}5$. Javob: **12 ta sinfga**.

---

17. Qavslarni ochib tekshiramiz: $-6 \cdot 2a = -12a$, $-6 \cdot (-3b) = +18b$ ✓.
Qolganlari: $-12a - 18b$, $12a - 18b$, $12a - 18b$. Javob: $-6(2a - 3b)$.

---

18. Juftlab: $2ax - 2ay = 2a(x - y)$, $3x - 3y = 3(x - y)$.
$2a(x - y) + 3(x - y) = (x - y)(2a + 3)$. Javob: $(x - y)(2a + 3)$.

---

19. $5a - 10b = 5(a - 2b)$.
$10a - 5b = 5(2a - b)$, $5a + 10b = 5(a + 2b)$. Javob: $5a - 10b$.

---

20. $S_A - S_B = 12xy + 3x - 10xy - 12x = 2xy - 9x = x(2y - 9)$.
$x > 0$, demak $2y - 9 = 0$, $y = 4{,}5$. Javob: **4,5**.
